

Chen Hongji

Male | Age: 27 years old | ☎ 15521372540 | ✉ lupangdewuyu@gmail.com
3 year work experience | Position: Full Stack/AI Engineer

Websites

<https://github.com/944750720>
<https://www.linkedin.com/in/hongji-chen/>

Education

The University of Tokyo	Master	EEIS	2022-2024
Hirose-Natsuaki Lab: Conducted research focused on Complex-valued Neural Networks and Radar Signal Processing; proficient in Python data processing libraries including NumPy and Pandas.			
South China University of Technology	Bachelor	Electrical Engineering and Its Automation	2016-2020

Professional Summary

- Fluent in Japanese(JLPT N1) and English, with long-term working experience in bilingual environments
- 2+ years of remote work experience, self-disciplined, proficient in asynchronous communication and collaboration
- Strong ownership and passion for technology, continuously learning AI application skills while emphasizing code quality and software engineering principles
- Proficient in both traditional full-stack development and AI Agent backend development

Skills

- **AI & Large Language Models:** LLMOps, Deterministic LLM Programming, LangChain, RAG, Cursor, Claude Code
- **Programming Languages:** Python, TypeScript
- **Frontend Development:** React.js, Next.js, Antd, Tailwind CSS, Linaria, Framer Motion, Jotai
- **Backend Development & DBs & :** FastAPI, GraphQL, PostgreSQL, Alembic, Drizzle ORM, SQLAlchemy, Atlas
- **Cloud, DevOps & Data:** AWS, GCP, Azure, Docker, Nginx, Terraform, GitHub Actions, dbt, Dagster
- **Testing & Quality:** unittest, Pytest, Vitest, Web Crawling, SEO
- **Languages:** Chinese (Native), English (Business Level), Japanese (Business Level - JLPT N1)

Employment History

Chemican Inc.	AI Software Engineer	2024.10-Now
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AI Backend Engineer (Full-time) | 2026.04 - Present

- **Core Feature Development for SDS Extraction Pipeline:** Led the development of multiple core features for the SDS (Safety Data Sheet) AI extraction pipeline, including cross-section regulatory mapping, company-specific rule override mechanisms, and auto-generation of annotator reference notes. Involved coordinated multi-file changes across extractors, schemas, prompts, and processing logic.
- **AI-Assisted Development Workflow & Large-Scale Refactoring:** Established an AI-assisted development loop centered on Code Review, defining a review checklist covering naming consistency, Pydantic usage boundaries, redundant comments, and type safety. Eliminated redundant LLM prompt logic and refactored cross-section

abstraction layers, reducing hundreds of lines of code while improving LLM context efficiency and model performance. Incrementally introduced pyright static type checking, significantly enhancing codebase maintainability.

AI Backend Engineer (Part-time) | 2025.04 – 2026.03

- **Data Access & API Development:** Independently designed and developed the API access layer for the SDS database using FastAPI and Google BigQuery, optimizing query and interaction efficiency for massive datasets.
- **Multilingual Prompt Engineering:** Designed multilingual LLM extraction prompts for Japanese/Chinese SDS documents. Combined Pydantic Schema constraints with Context Engineering to support structured extraction across 15 SDS sections, improving extraction accuracy.
- **Automation & Containerized Deployment:** Utilized Terraform, Nginx, and Google Cloud Run to implement the containerized deployment of dbt data analysis results (static HTML pages), ensuring high availability and rapid iteration of data visualization dashboards.

AI Backend Engineer Intern | 2024.10 – 2025.04

- **Architectural Innovation:** Spearheaded the company's first integration of Schema mechanisms to constrain LLM formatted outputs, filling a critical technical gap in standardized LLM generation for the team.
- **Technical Breakthrough:** Played a pivotal role in LLM backend architecture development; designed and implemented an error-feedback retry mechanism combining Pydantic validation and context engineering to empower the LLM with closed-loop self-correction capabilities.
- **Business Impact:** Eliminated intermittent parsing errors, drastically improving the accuracy and stability of structured data outputs for LLM-powered OCR tasks.

pluszero, Inc Full Stack Engineer

2025.04-2026.04

Enterprise Management System Frontend Development

- **Core Module Engineering:** Engineered core functional modules and permission management logic for an enterprise-level management system utilizing React.
- **GraphQL Integration:** Integrated with backend GraphQL interfaces to complete data interaction between the frontend and backend.

Enterprise Chatbot Demo Development

- **End-to-End Architecture:** Utilized the FastAPI + React tech stack to build the complete interaction process from database design, covering backend API design and frontend component development.

IGSA Inc. AI Software Engineer Intern

2024.10-2025.03

Built an enterprise-grade AI chatbot using RAG and LangChain, powered by an internal error code resolution knowledge base, to automatically answer users' technical questions.

- **Hybrid Retrieval Strategy:** Combined keyword search (BM25) with vector retrieval (Embedding), significantly improving retrieval effectiveness.
- **Quantified Impact:** Achieved over 90% Recall@10 for error code retrieval, enabling the chatbot to locate relevant solutions with greater accuracy and coverage.

Kuzen Inc. AI Software Engineer Intern

2023.11-2024.10

- **Technical Implementation:** Expanded LLM knowledge base coverage by engineering technical solutions to crawl dynamic websites, PDF files, and text files as authoritative sources.
- **Automated Pipeline:** Automated data pipelines utilizing AWS S3, Lambda, and EventBridge to execute weekly crawling and registration of internal Notion content. Enhanced the chatbot's ability to accurately answer user questions regarding product specifications and user guides.

Project List

Blog

A personal blog site built with Next.js, Tailwind CSS, and Framer Motion, deployed at hongji.xyz, used for technical sharing and showcasing personal projects.